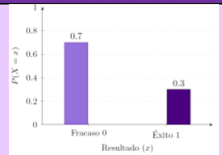
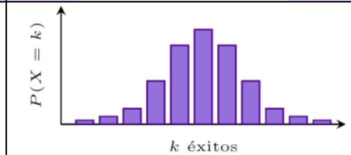
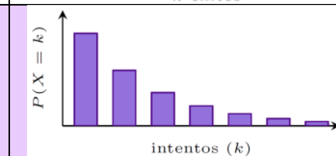
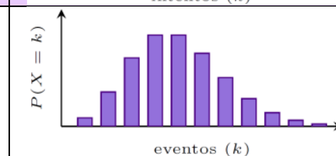
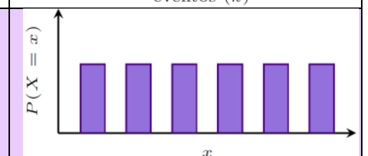
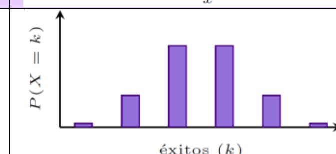
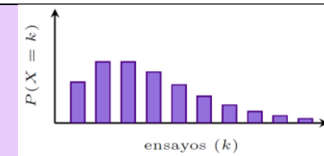
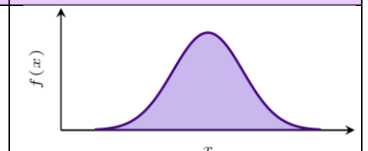
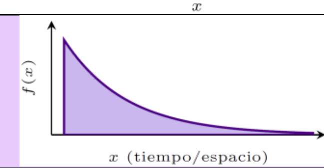
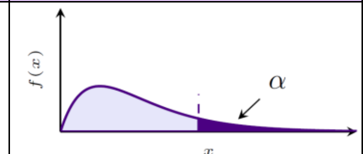
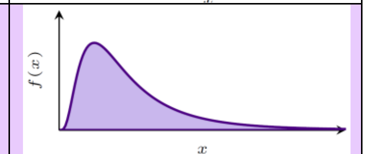
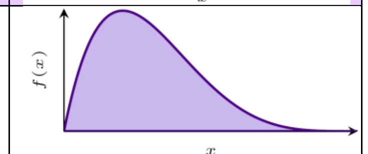
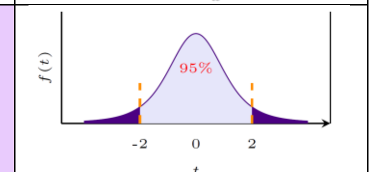
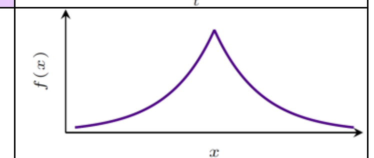
Tipo	Nombre y aplicación	Nomenclatura	Dominio	Parámetros	Función de masa o densidad	Función acumulativa	Media $E[x]$	Varianza Var[x]	Desv. Est. σ	F.G.M. $M_X(t)$	Gráfica
Discreta	<u>Bernoulli</u> Experimento con éxito o fracaso.	$X \sim Ber(p)$	$\{0,1\}$	$0 < p < 1$	Masa $P(X=x) = p^x(1-p)^{1-x}$	$F(x) = \begin{cases} 0 & x < 0 \\ 1-p & 0 \le x < 1 \\ 1 & x \ge 1 \end{cases}$	p	$p(1-p)$	$\sqrt{p(1-p)}$	$(1-p)+pe^t$	
Discreta	***<u>Binomial</u>*** *Número de éxitos en n ensayos.*	$X \sim Bin(n,p)$	$k = 0,1,\dots,n$	$n \in \mathbb{N}$ $0 < p < 1$	Masa $P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$	$F(k) = \sum_{i=0}^{k} \binom{n}{i} p^i (1-p)^{n-i}$	np	$np(1-p)$	$\sqrt{np(1-p)}$	$(1-p+pe^t)^n$	
Discreta	***<u>Geométrica</u>*** *Ensayos hasta el primer éxito.*	$X \sim Geo(p)$	$k = 1,2,\dots$	$0 < p < 1$	*Masa* $P(X=k) = p(1-p)^{k-1}$	$F(k) = 1 - p(1-p)^k$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\sqrt{\frac{1-p}{p^2}}$	$\frac{pe^t}{1-(1-p)e^t}$	
Discreta	***<u>Poisson</u>*** *Conteo de eventos raros.*	$X \sim Pois(\lambda)$	$k = 0,1,2,\dots$	$\lambda > 0$	*Masa* $P(X=k) = \frac{e^{-\lambda}\lambda^k}{k!}$	$F(k) = \sum_{i=0}^{k} \frac{e^{-\lambda}\lambda^i}{i!}$	λ	λ	$\sqrt{\lambda}$	$e^{\lambda(e^t-1)}$	
Discreta	***<u>Uniforme discreta</u>*** *Lanzamiento de un dado.*	$X \sim UD(a,b)$	$x \in \{a, a+1, \dots, b\}$	$a, b \in \mathbb{Z}$ $a < b$ $n = b - a + 1$	*Masa* $P(X=x) = \frac{1}{n}$	$F(x) = \frac{\lfloor x \rfloor - a + 1}{n}$	$\frac{a+b}{2}$	$\frac{n^2-1}{12}$	$\sqrt{\frac{n^2-1}{12}}$	$\frac{e^{at} - e^{(b+1)t}}{n(1-e^t)}$	
Discreta	***<u>Hipergeométrica</u>*** *Muestreo sin remplazo*	$X \sim HG(N,K,n)$	$x \in \{\text{máx}(0, n-(N-K)), \dots, \text{mín}(n,K)\}$	*N=población* *K=éxitos en N* *N=muestra*	*Masa* $P(X=x) = \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$	$F(x) = P(X \le x) = \sum_{i=0}^{x} \frac{\binom{K}{i}\binom{N-K}{n-i}}{\binom{N}{n}}$	$n\frac{K}{N}$	$n\frac{K}{N}\frac{N-K}{N}\frac{N-n}{N-1}$	$\sqrt{n\frac{K}{N}\frac{N-K}{N}\frac{N-n}{N-1}}$	$\sum_{x=0}^{n} e^{tx} \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}$ *M=éxitos poblacionales*	
Discreta	***<u>Binomial negativa/Pascal</u>*** *Pascal: cuenta el número total de ensayos para r éxitos* *Binomial neg.: cuenta los fracasos*	$X \sim BN(r,p)$	$x = r, r+1, r+2, \dots$	$r \in \mathbb{N}$ *Éxitos deseados* $0 < p < 1$	*Masa* $P(X=x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}$	$F(k;r,p) = \sum_{i=0}^{k} \binom{i+r-1}{i} p^r (1-p)^i$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$	$\sqrt{\frac{r(1-p)}{p^2}}$	$\left(\frac{pe^t}{1-(1-p)e^t}\right)^r$	
Continua	***<u>Normal</u>*** *Fenómenos naturales*	$X \sim N(\mu, \sigma^2)$	$(-\infty, \infty)$	$\mu \in \mathbb{R}$ $\sigma > 0$	*Densidad* $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$F(x) = \int_{-\infty}^{x} f(t)dt$	μ	σ^2	σ	$e^{\mu t + \frac{1}{2}\sigma^2 t^2}$	
Continua	***<u>Exponencial</u>*** *Tiempo de espera.*	$X \sim Exp(\lambda)$	$x \ge 0$	$\lambda > 0$	*Densidad* $f(x) = \lambda e^{-\lambda x}$	$F(x) = 1 - e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{1}{\lambda}$	$\frac{\lambda}{\lambda - t}$	

Continua	**Uniforme** Eventos equiprobables.	$X \sim U(a,b)$	$a \le x \le b$	$a < b$	Densidad $f(x) = \frac{1}{b-a}$	$F(x) = \frac{x-a}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{b-a}{\sqrt{12}}$	$\frac{e^{tb}-e^{ta}}{t(b-a)}$	
Continua	**Gamma**	$X \sim \Gamma(\alpha,\beta)$	$x > 0$	Forma $\alpha > 0$ Escala $\beta > 0$	Densidad $f(x) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}$	$F(x) = \frac{\gamma(\alpha, x/\beta)}{\Gamma(\alpha)}$	$\alpha\beta$	$\alpha\beta^2$	$\sqrt{\alpha\beta^2}$	$(1-\beta t)^{-\alpha}$ Para $t < \frac{1}{\beta}$	
Continua	**Ji-Cuadrada**	$X \sim \chi^2(k)$	$x \ge 0$	$k \in \mathbb{N}$ Grados de libertad	Densidad Es una Gamma con $\alpha = \frac{k}{2}$ y $\beta = 2$	$F(x) = \frac{\gamma\left(\frac{k}{2}, \frac{x}{2}\right)}{\Gamma\left(\frac{k}{2}\right)}$	k	$2k$	$\sqrt{2k}$	$(1-2t)^{-\frac{k}{2}}$	
Continua	**Lognormal**	$X \sim LN(\mu\sigma^2)$	$X > 0$	$\mu =$ media (ubicación) $\sigma =$ desviación estandar (escala)	Densidad $f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x-\mu)^2}{2\sigma^2}}$	$F(x;\mu\sigma) = \Phi\left(\frac{\ln(x)-\mu}{\sigma}\right)$	$e^{\mu+\frac{\sigma^2}{2}}$	$(e^{\sigma^2}-1)e^{2\mu+\sigma^2}$	$\sqrt{(e^{\sigma^2}-1)e^{2\mu+\sigma^2}}$	No tiene una F.G.M. definida	
Continua	**Beta**	$X \sim B(\alpha\beta)$	$0 \le x \le 1$	$\alpha\beta > 0$	Densidad $f(x) = \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1}$	$F(x;\alpha\beta) = I_x(\alpha\beta) = \frac{\int_0^x t^{\alpha-1}(1-t)^{\beta-1}dt}{B(\alpha\beta)}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$\sqrt{\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}}$	$1 + \sum_{n=1}^{\infty} \frac{B(\alpha+n\beta)}{B(\alpha\beta)} \frac{t^n}{n!}$	
Continua	**t de Student**	$T \sim t_v$	$(-\infty,\infty)$	T=variable aleatoria v=grados de libertad (n-1)	Densidad $f(x) = \frac{\Gamma\left(\frac{v+1}{2}\right)}{\sqrt{v\pi}\,\Gamma\left(\frac{v}{2}\right)} \left(1+\frac{t^2}{v}\right)^{-\frac{v+1}{2}}$	$F(x \setminus v) = P(T \le x) = \int_{-\infty}^{x} f(u \setminus v)du$	0 si v>1	$\frac{v}{v-2}$ si $v > 2$	$\sqrt{\frac{v}{v-2}}$ si $v > 2$	$\int_{-\infty}^{\infty} e^{tx} f(x)dx$	
Continua	**Doble exponencial**	$X \sim Laplace(\mu,b)$	$-\infty < x < \infty$	$\mu =$ ubicación $b =$ escala	Densidad $f(x) = \frac{1}{2b} e^{-\frac{\lvert x-\mu \rvert}{b}}$	$F(x) = \begin{cases} \frac{1}{2} e^{\frac{x-\mu}{b}} & si\ x < \mu \\ 1 - \frac{1}{2} e^{-\frac{x-\mu}{b}} & si\ x \ge \mu \end{cases}$	μ	$2b^2$	$\sqrt{2b^2}$	$\frac{e^{\mu t}}{1-b^2t^2}$	
Continua	**Cauchy**	$X \sim Cauchy(x_0\gamma)$	$-\infty < x < \infty$	$x_0 =$ ubicación $\gamma =$ escala	Densidad $f(x) = \frac{1}{\pi\gamma\left[1+\left(\frac{x-x_0}{\gamma}\right)^2\right]}$	$F(x;x_0\gamma) = \frac{1}{\pi} arctan\left(\frac{x-x_0}{\gamma}\right) + \frac{1}{2}$	No tiene	No tiene	No tiene	No tiene